Contrôle PID

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Introduction

- Contrôle/asservissement
- Modélisation mathématique
- Systèmes
  - Transformation Laplace
  - Fonctions de transfert
- Contrôle PID

Contrôle

Principes des systèmes complexes

- Concepts essentiels
  - Stabilité/Sensibilité (1890s)
  - Observabilité – Contrôlabilité/régulabilité (1950s)
  - Robustesse (1970s)

Exemple de contrôle

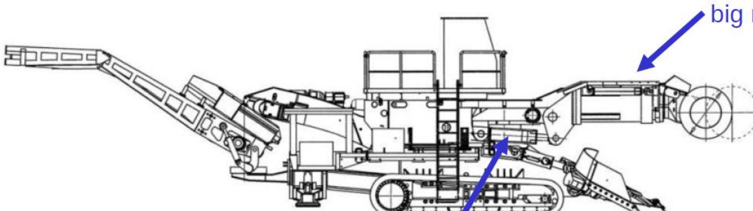
How do we get the cutting arm of this machine to move to a certain point?

Courtesy of Sandvik Mining and Construction, 2011

What is being controlled?

- « Processus »

Here we have a big mass



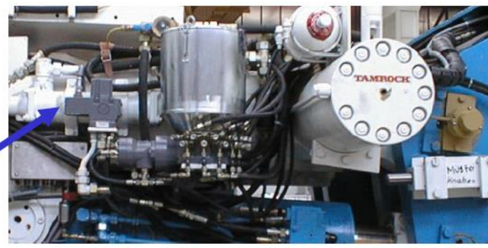
Courtesy of Sandvik Mining and Construction, 2011

A hydraulic cylinder is used to move the mass

Capteurs et actionneurs

- On doit savoir la position
- L'actionneur initie le mouvement
- L'actionneur est la valve hydraulique, qui prend un tension comme signal (la variable contrôlé)

A hydraulic proportional valve will move the cylinder

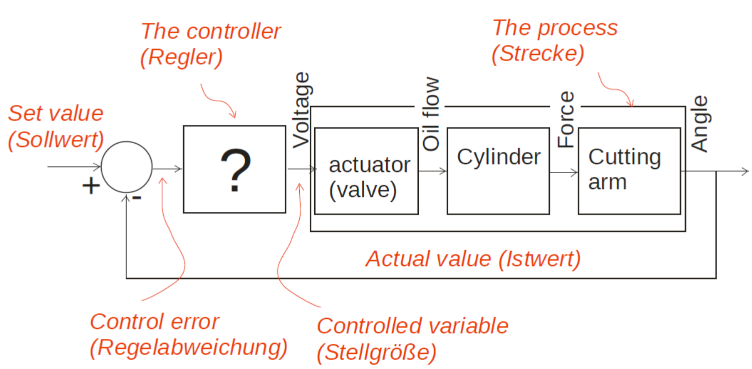


This is a position (angle) sensor

Courtesy of Sandvik Mining and Construction, 1998

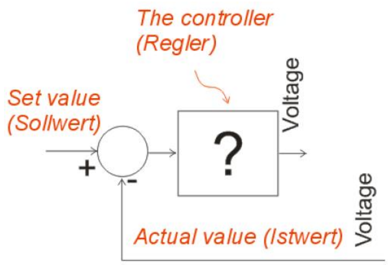
Circuit de réglage de base

- Signal, combinaison, processus

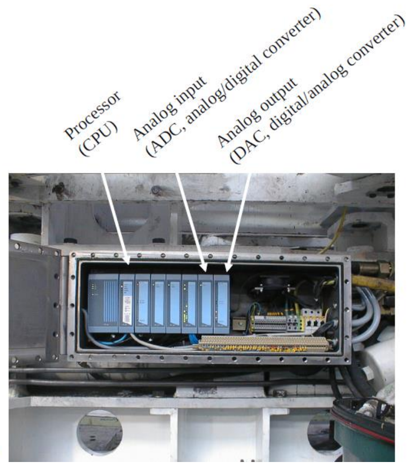


Contrôleur

- Avec la processus donné, comment bâtir un contrôleur



Of course it is a piece of
software on some
kind of computer!



© 2002, Gerhard Rath

L’emploi de l’ingénieur de contrôle

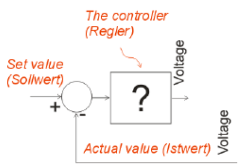
Oh, I will go to the machine, when it is operable, and I shall have
a lot of time for programming and testing!
Maybe in the night, when everything is quiet!

The ugly truth: Maybe you see your plant the first time when it should start to work!
Consequence: Test as early as possible! We need:

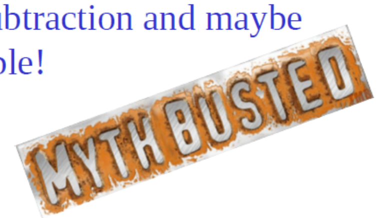
Simulation models:
Numerical solutions & tools!

Mathematical models:
Differential Equations, Laplace Transform!

Comment est-ce que ça va fonctionner ?



Oh, I look sometimes on the actual value, and then I correct a little bit! A simple subtraction and maybe a multiplication or a look-up table!



The ugly truth: Control loops tend to be unstable; The customer never is satified with the accuracy!

Consequence:

Control Theory:

Design Methods, Stability analysis, Structure of controllers, …

Dynamique

Equations of motion are the cornerstone

Newton's law for transitional motion states

$$F = \frac{\mathrm{d}}{\mathrm{d}t}(mv)$$

Where: F is the vector sum of all forces applied to each body in the system and has the dimension Newtons [N], m is the mass of the body in kilograms [kg] and v is the velocity in meters per second [m/sec].

Note 1: The is formulated with respect to the change in momentum. In most cases the mass is assumes to be constant and the law can be expressed in its more common form.

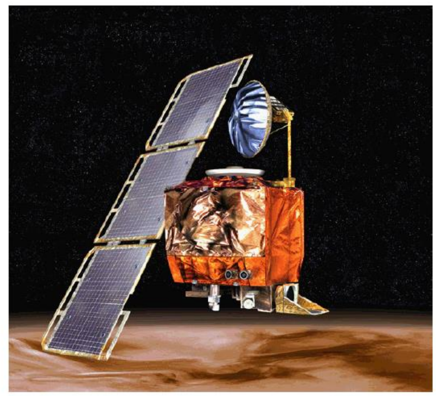
$$F = m\ddot{x} = ma$$

Note 2: All equations are formulated with respect to a fixed inertial frame, the fixed stars (not the earth since it is in motion): This has little practical consequences however, it has interesting philosophical connotations.

Mouvement rotationnel

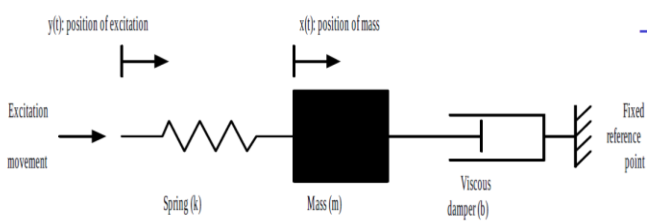
- Newton's law applied to rotary motion gives:
  - M is the sum of all moments working on the center of mass of the body [Nm]
  - I is the body's moment of inertia [kg m2]
  - a is the angular acceleration of the body [rad/sec2]
- The same comments are true here as for the translational motion, the original equation is formulated with respect to the rate of change of the rotational inertia.
- Note: For many bodies e.g. aircraft, the mass changes during motion due to fuel consumption.

$$I\ddot{\varphi} = \sum M$$



Modèle mécanique

- Position input to spring mass damper system



- Newton's law

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + ky(t)$$

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = ky(t)$$

Modèle d'un moteur CC

- Law of motors (B magnetic flux density, l length of conductor)

$$F = Bli \quad \text{[newtons]}$$

- Torque as a function of armature current

$$T = K_M i_a$$

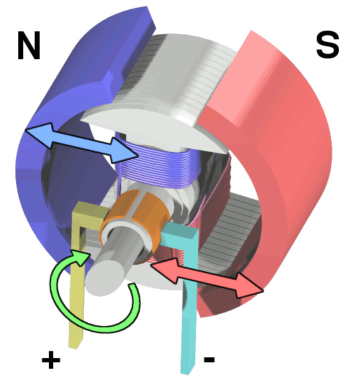
- Law of generators (v velocity of conductor)

$$e(t) = Blv \quad \text{[volts]}$$

- Back EMF (Counter EMF)

$$u_{\text{emf}} = K_e \omega$$

K_M = Torque constant
K_e = Back emf constant
i_a = Armature current
ω = rotation speed



[Wikipedia]

Rappel : Transforme de Laplace

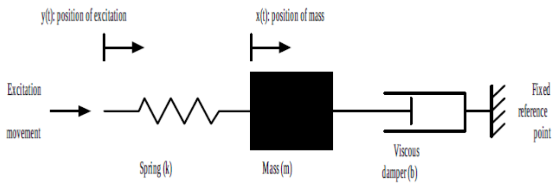
Laplace Transform:

$$\mathcal{L}\{f(t)\} = F(s) \triangleq \int_{0^-}^{\infty} f(t)\mathrm{e}^{-st}\mathrm{d}t$$

Inverse Laplace Transform:

$$\mathcal{L}^{-1}\{F(s)\} = f(t) \triangleq \frac{1}{2\pi \mathrm{j}} \int_{0-\mathrm{j}\infty}^{+\mathrm{j}\infty} F(s)\mathrm{e}^{st}\mathrm{d}s$$

Exemple



- The differential equation of motion for this system is:

$$m\frac{\mathrm{d}^2x(t)}{\mathrm{d}t^2} + b\frac{\mathrm{d}x(t)}{\mathrm{d}t} + kx(t) = ky(t)$$

  – Where b is the coefficient of viscous damping with the dimension [N/m/sec] and k is the Hook's constant of the spring [N/m].

- The equation can be more briefly written as:

$$m\ddot{x} + b\dot{x} + kx = ky$$

  – This is a linear time invariant differential equation if the coefficients i.e. m, k and b do not change.

Solution par transformation de Laplace

$$\boxed{X(s) = \frac{kY(s)}{ms^2 + bs + k} + \frac{x(0)(sm + b)}{ms^2 + bs + k} + \frac{m\dot{x}(0)}{ms^2 + bs + k}}$$

In the previous lecture we got *x(t)* with the inverse transform. This is not interesting here!

We look at the solution for *X(s)* and find:

1. The first term is a contribution of the input function!
1. Each of the initial conditions contribute in separate terms!
2. Every term has the same nominator!

The solution is a superposition of the response to the input function and of any of the initial conditions!

Fonction de transfert

If we set all initial conditions equal zero, we get $X(s) = H(s)\, Y(s)$
with *H(s)* is the ***transfer function***.

It is the quotient of the output function by the input function:

$$\boxed{H(s) = \frac{\overset{\text{output}}{X(s)}}{\underset{\text{input}}{Y(s)}}}$$

In our example, the transfer function is: $H(s) = \dfrac{X(s)}{Y(s)} = \dfrac{k}{ms^2 + bs + k}$

Utilisation des fonctions de transfert

What control engineers do for modeling is:
Start the system with all initial conditions zero (rest position, Ruhelage)

The consequences are:
Every mass has its own coordinate system starting at zero.
We cannot treat initial conditions with transfer functions.

Avantages des fonctions de transfert (I)

Why are we so happy with the transfer function H(s)? $X(s) = H(s)\,Y(s)$

1 We can calculate the output response easily:
For example, if the input is a step function, Y(s) = 1/s, then the output is: $X(s) = H(s)\,\frac{1}{s}$

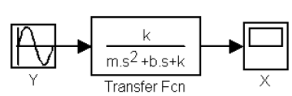
a) We can easily see where the system goes after a long time with the ***limit theorem!***

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s)$$

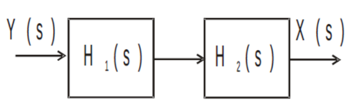
b) We can investigate the ***stability*** of a system (see later)

Avantages des fonctions de transfert (II)

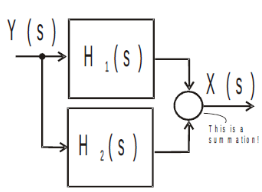
2 Every numerical simulation tool accepts transfer functions as a description for dynamic systems:



3 We find transfer functions of complex systems:



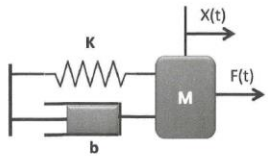
$$\frac{X(s)}{Y(s)} = H_1(s) \cdot H_2(s)$$



$$\frac{X(s)}{Y(s)} = H_1(s) + H_2(s)$$

This follows directly from the principle of superposition!

Système d'ordre 2



$$F(t) = ma + bV + Kx$$

$$F(t) = m\frac{d^2x}{dt} + b\frac{dx}{dt} + Kx$$

Dans le domaine de Laplace

$$F(s) = s^2 mx(s) + bsx(s) + Kx(s)$$

$$F(s) = x(s)\left(ms^2 + bs + K\right)$$

$$\frac{1}{ms^2 + bs + K} = \frac{x(s)}{F(s)}$$

$$\frac{1/m}{s^2 + \frac{b}{m}s + \frac{K}{m}} = G(s)$$

Forme générale

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

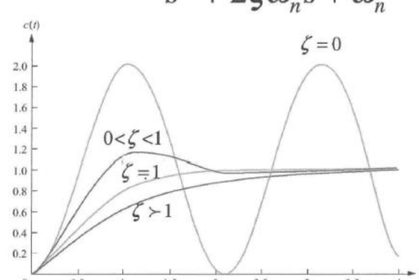
ω_n Fréquence naturelle:
Fréquence de l'oscillation sans amortissement

ζ Amortissement:
Temps nécessaire pour atténuer les oscillations

Réponse

ORDRE 2

$$G(s) = K_s \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



$\zeta = 0 \quad \Rightarrow$ *système – NON – AMORTI*

$0<\zeta<1 \quad \Rightarrow$ *système – SOUS – AMORTI*

$\zeta = 1 \quad \Rightarrow$ *système AMORTI – CRITIQUE*

$\zeta \succ 1 \quad \Rightarrow$ *système SUR – AMORTI*

Model of the Process to be Controlled

Analysis of controllers will be done with a process (plant, Regelstrecke) of second order:

$$H(s) = V \cdot \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$H(s) = V \cdot \frac{1}{\frac{s^2}{\omega_n^2} + \frac{2\zeta}{\omega_n} s + 1}$$

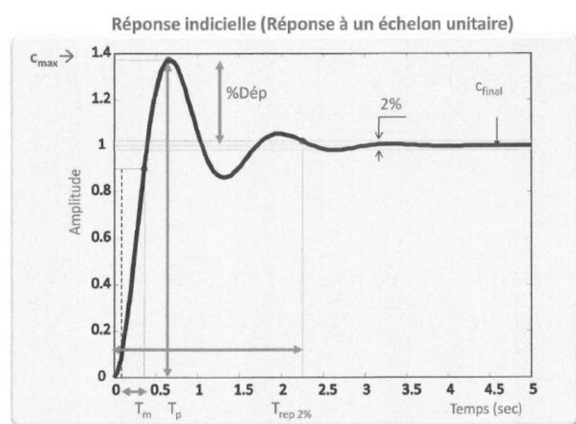
with:

ω_n ... natural frequency, Eigen frequency (Eigenfrequenz)

ζ ... damping ratio (Dämpfungsgrad)

V ... gain (Verstärkung)

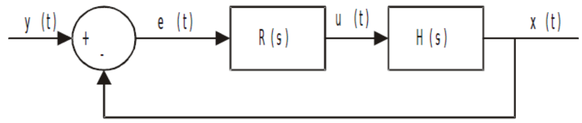
Réponse, ordre 2



Tp: temps mis pour atteindre le 1er maximum.(Peak time)
Tmontée: temps requis pour passer de 10 à 90 % de la valeur finale en régime permanent
Trep: temps de réponse. le temps mis pour se situer à ± T% de la valeur en régime permanent
%Dép: la quantité de dépassement relative à la valeur du signal au temps Tp. Cette valeur est exprimée en % de la valeur en régime

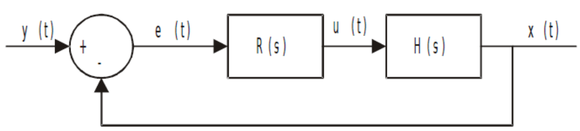
Basic Controller Structure

A possible structure to bring the output of the process H(s) to a desired value is:



With:
y … set value (Sollwert)
x … actual value (Istwert)
e … control error (Regelabweichung)
u … controlled variable (Stellgröße)
H(s) … process (Regelstrecke)
R(s) … controller (Regler)

Fonction de transfert, boucle fermé



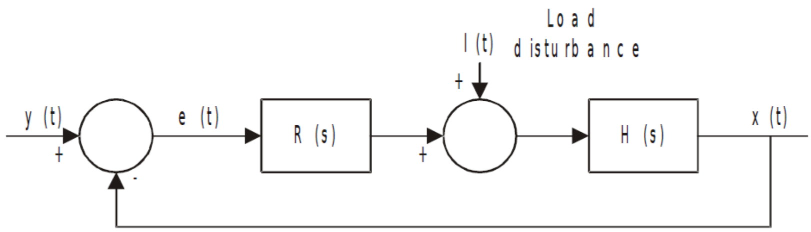
$$X(s) = R(s)\,H(s)\,E(s)$$
$$= RH(Y - X)$$

$$X(1 + RH) = RHY$$

$$\boxed{T_{IO}(s) = \frac{X(s)}{Y(s)} = \frac{R(s)\,H(s)}{1 + R(s)\,H(s)}}$$

Input to output transfer function,
Führungsübertragungsfunktion

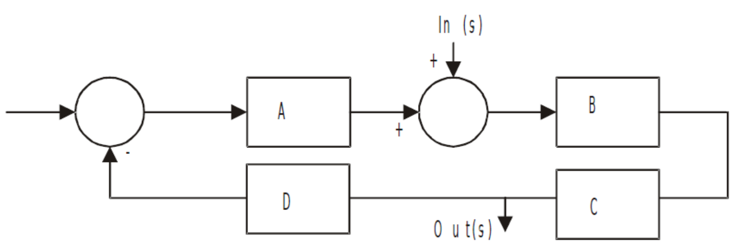
Perturbation



$$\boxed{T_{LO}(s)=\frac{H(s)}{1+R(s)H(s)}}$$

Load to output transfer function,
Störübertragungsfunktion

Fonction de transfert, par inspection



Numerator: Product of blocks on the path from input to output;
Denominator: 1 + Product of blocks in the loop;

$$\frac{Out(s)}{In(s)}=\frac{BC}{1+ABCD}$$

Pourquoi rétroaction ?

- Enable the system to attain an acceptable steady state error
- Enable the system to attain acceptable tracking error
- Reduce system sensitivity to variations in system parameters
- Control rise time, overshoot, settling time, etc. of step response
- Control an unstable system through a closed control loop

Feedback (Rétroaction)

- The return to the input of a part of the output of a machine, system, or process (as for producing changes in an electronic circuit that improve performance or in an automatic control device that provide self-corrective action) [Merriam Webster Dictionary]
- Watt's Steam Engine Flyball Governor

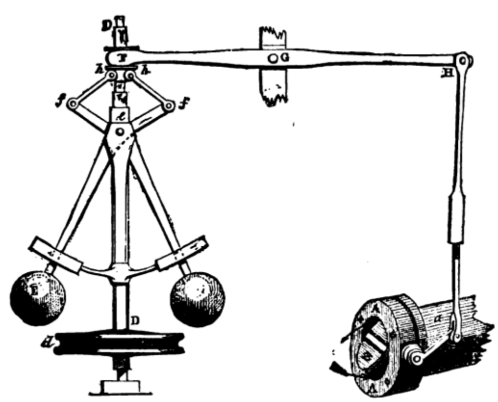


FIG. 4.—*Governor and Throttle-Valve.*

Cruise control

- Regular and adaptive

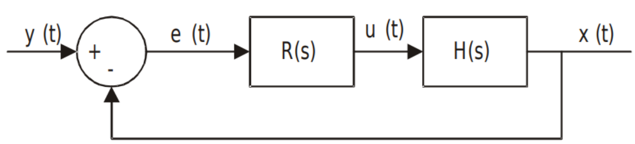
[Wikipedia]

[Wikipedia]

P (Proportional) Control

The output is simply proportional to the control error:

$$u(t) = k_p \{y(t) - x(t)\}$$
$$U(s) = k_p \{Y(s) - X(s)\}$$



The transfer function is: $T_{IO}(s) = \dfrac{RH}{1+RH}$

With $R(s) = k_p$ and $H(s) = V \cdot \dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

follows:

$$T_{IO} = \frac{V k_p \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2 (1 + V k_p)}$$

Eigenfrequency and Damping (P Control)

The transfer function of the control loop is of 2nd order; now $\omega_{n,P}$
Can be defined as the natural frequency and ζ_P as the damping factor of the P controlled system:

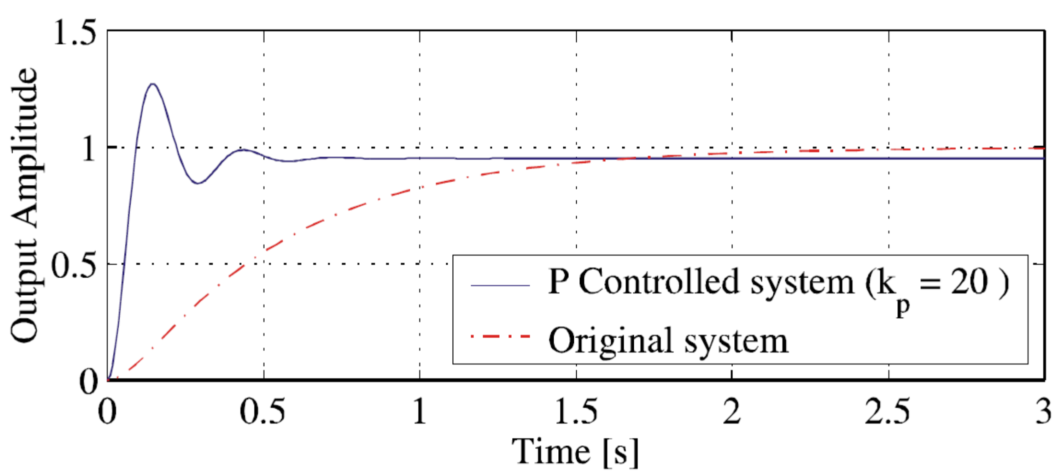
$$T_{IO} = \frac{V k_p \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2 \left(1 + V k_p\right)} \equiv \frac{\omega_{n,P}^2}{s^2 + 2\zeta_P \omega_{n,P} s + \omega_{n,P}^2}$$

Comparing the coefficients gives:

$$\omega_{n,P} = \omega_n \sqrt{1 + V k_p} \qquad \zeta_P = \frac{\zeta}{\sqrt{1 + V k_p}}$$

Conclusions: The proportional gain kp *increases the natural frequency* and *decreases the damping*, this means, it makes the system faster, but increases the tendency to oscillate!

Contrôleur P, réponse à échelon unité



Steady State Error

$$T_{IO} = \frac{Vk_p\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2\left(1 + Vk_p\right)}$$

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

$$E(s) = Y(s) - X(s) = Y(s)\left(1 - \frac{X(s}{Y(s)}\right) = Y(s)\left(1 - T_{IO}(s)\right)$$

$$Y(s) = \frac{1}{s}$$

We choose the step function as input!

$$\lim_{s\to 0} sE(s) = \frac{s}{s}\left(\frac{s^2 + 2\zeta\omega_n s + \omega_n^2\left(1 + Vk_p\right) - Vk_p\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2\left(1 + Vk_p\right)}\right) = \frac{1}{1 + Vk_p}$$

$$\boxed{e_\infty = \frac{1}{1 + Vk_p}}$$

Increasing k_p reduces the steady state error!

Limitations for k_p

The damping factor of the closed loop is:

$$\zeta_P = \frac{\zeta}{\sqrt{1 + Vk_p}}$$

For a design without overshoot ζ_P=1 follows:

$$k_{p,max} = \frac{1}{V}\left(\zeta^2 - 1\right)$$

For reasonable good damping $\zeta_P = 2^{-1/2} = 0.7$ follows:

$$k_{p,max} = \frac{1}{V}\left(2\zeta^2 - 1\right)$$

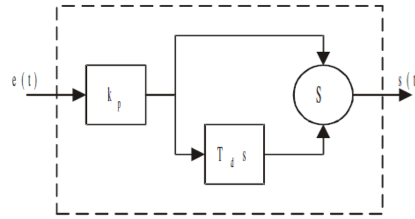
PD (Proportional-Derivative) Control

Takes the *rate of change* into account:

$$u(t) = k_p \{e(t) + T_d \dot{e}(t)\}$$

$$U(s) = k_p \{E(s) + T_d s E(s)\}$$

$$\frac{U(s)}{E(s)} = R_{PD}(s) = k_p \{1 + T_d s\}$$



$$\boxed{R_{PD}(s) = k_p \{1 + T_d s\}}$$

With the process $H(s) = V \cdot \dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ follows the loop transfer function:

$$T(s) = \frac{RH}{1+RH} = \frac{V k_p (1 + T_d s) \omega_n^2}{s^2 + (V k_p \omega_n^2 T_d + 2\zeta\omega_n) s + \omega_n^2 (1 + V k_p)}$$

Properties of PD Control, 2nd Order System I

$$T(s) = \frac{V k_p (1 + T_d s) \omega_n^2}{s^2 + (V k_p \omega_n^2 T_d + 2\zeta\omega_n) s + \omega_n^2 (1 + V k_p)}$$

The loop now has poles and zeros; it is still of 2nd order; modeling the denominator with

$$s^2 + 2\zeta_{PD}\omega_{n,PD} s + \omega_{n,PD}^2$$

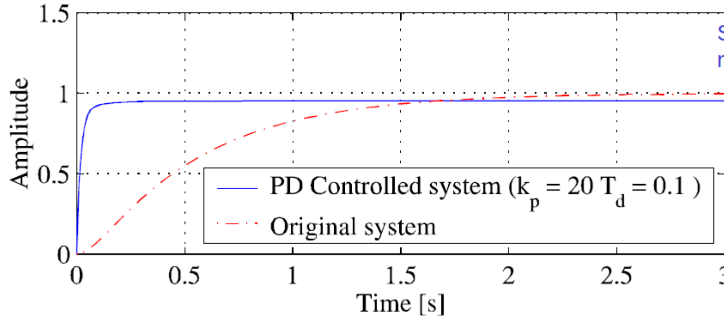
and comparing the coefficients gives:

$$\omega_{n,PD} = \omega_n \sqrt{1 + V k_p} = \omega_{n,P}$$ This is the same result as for P control!

$$\zeta_{PD} = \frac{\zeta}{\sqrt{1 + V k_p}} + \frac{T_d \omega_n V k_p}{2\sqrt{1 + V k_p}} = \zeta_P + \frac{T_d \omega_n V k_p}{2\sqrt{1 + V k_p}}$$

This means, the D portion increases the damping!

Contrôleur PD, système 2e ordre



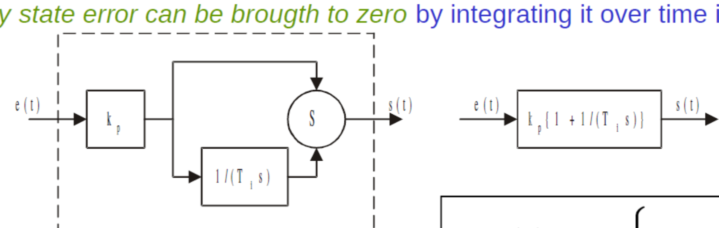
Step response;
mind the reduced overshoot!

The solution for the steady state error in PD is the same as for P control!

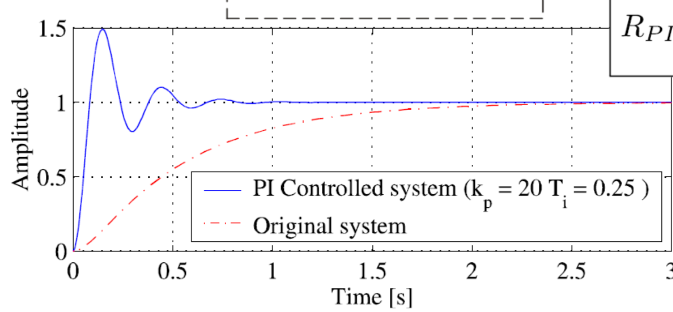
$$\boxed{e_\infty = \frac{1}{1 + V k_p}}$$

Contrôleur PI, système 2e ordre

The *steady state error can be brougth to zero* by integrating it over time in the controller.



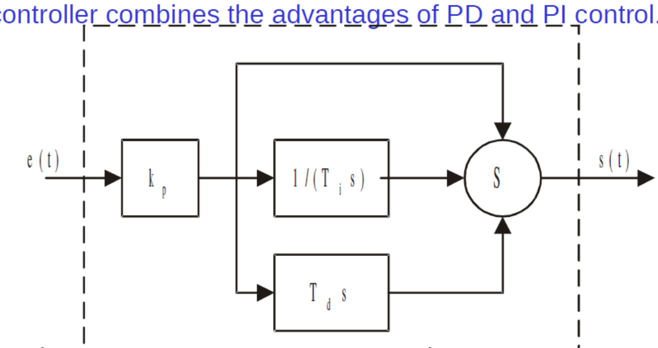
$$\boxed{R_{PI}(s) = k_p \left\{ 1 + \frac{1}{T_I s} \right\}}$$



$$\boxed{e_\infty = 0}$$

Contrôleur PID

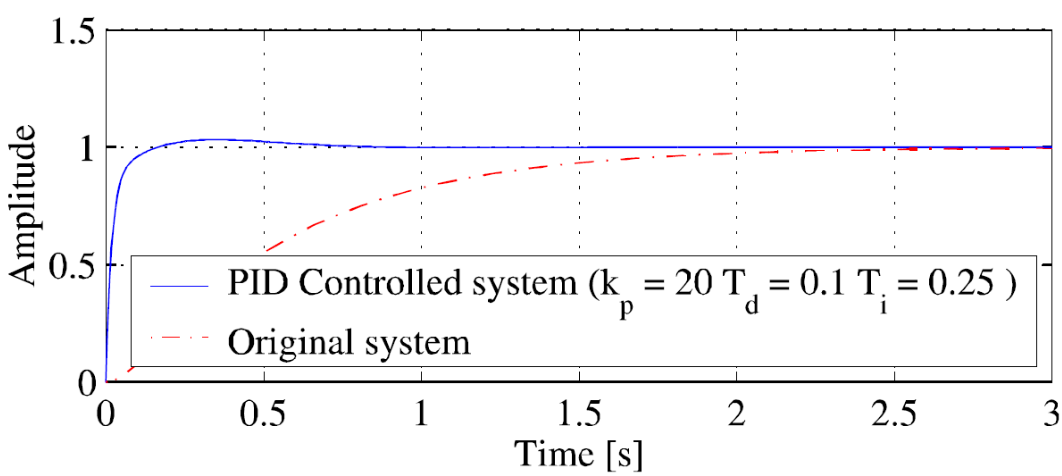
The PID controller combines the advantages of PD and PI control.



$$u(t) = k_p \left\{ e(t) + T_d \dot{e}(t) + \frac{1}{T_I} \int e(t) \mathrm{d}t \right\}$$

$$\frac{U(s)}{E(s)} = R_{PID}(s) \qquad \boxed{R_{PID}(s) = k_p \left\{ 1 + T_d s + \frac{1}{s T_I} \right\}}$$

Réponse à échelon unité, PID



Paramètres PID

$$\boxed{R_{PID}(s) = k_p \left\{ 1 + T_d s + \frac{1}{sT_I} \right\}}$$

kp	Proportional Gain	Verstärkungsfaktor
T_I	Integral Reset Time	Nachstellzeit
T_D	Derivative Action Time	Vorhaltezeit

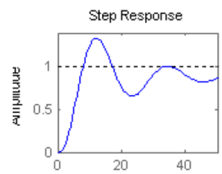
PID Tuning

Controller design means to find paramters for a desired behaviour of the loop. The most important methods are:

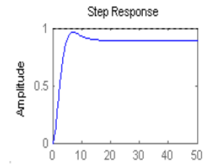
1 Root locus design
2 Ziegler-Nichols, method I
3 Zielger-Nichols, method II
4 Bode plot
5 Experimental design

Experimental: Step Response of Closed Loop

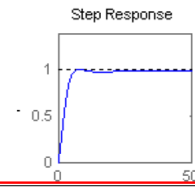
1. Adjust the proportional gain until reasonable overshoot (20%)



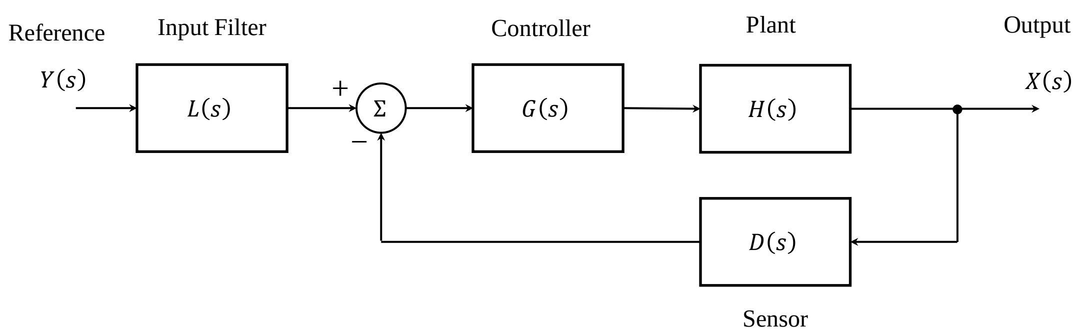
2. Adjust the derivative part to acceptable transient response and repeat from step 1 iteratively if necessary



3. Optimise integral part to remove steady state error



Système contrôlé



Spécifications (ordre 2)

Sous-amortie $0 < \zeta < 1$

$$\%Dép = \frac{c_{\max} - c_{final}}{c_{final}} \times 100$$

$$c_{\max} = 1 + e^{-\left(\zeta\pi / \sqrt{1-\zeta^2}\right)}$$

$$T_{rép.2\%} \approx \frac{4}{\zeta\omega_n} \quad T_{rép.5\%} \approx \frac{3}{\zeta\omega_n}$$

$$T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$$

$$\%Dép = e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}\right)} \times 100$$

$$\zeta = \frac{-\ln(Dép)}{\sqrt{\pi^2 + \ln^2(Dép)}}$$

Sur-amortie $\zeta > 1$

$$T_{rép.2\%} \approx \frac{4}{\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}}$$

$$T_{rép.5\%} \approx \frac{3}{\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}}$$

$$AKs = Valeur(F)$$

Sommaire

- Proportional control is a basic form of feedback control
- The derivative term in PD control modifies the damping of the closed loop system
- The integral term in PI control modifies the final value of the closed loop system
- The combined action (PID) can be used to effect a desired system response